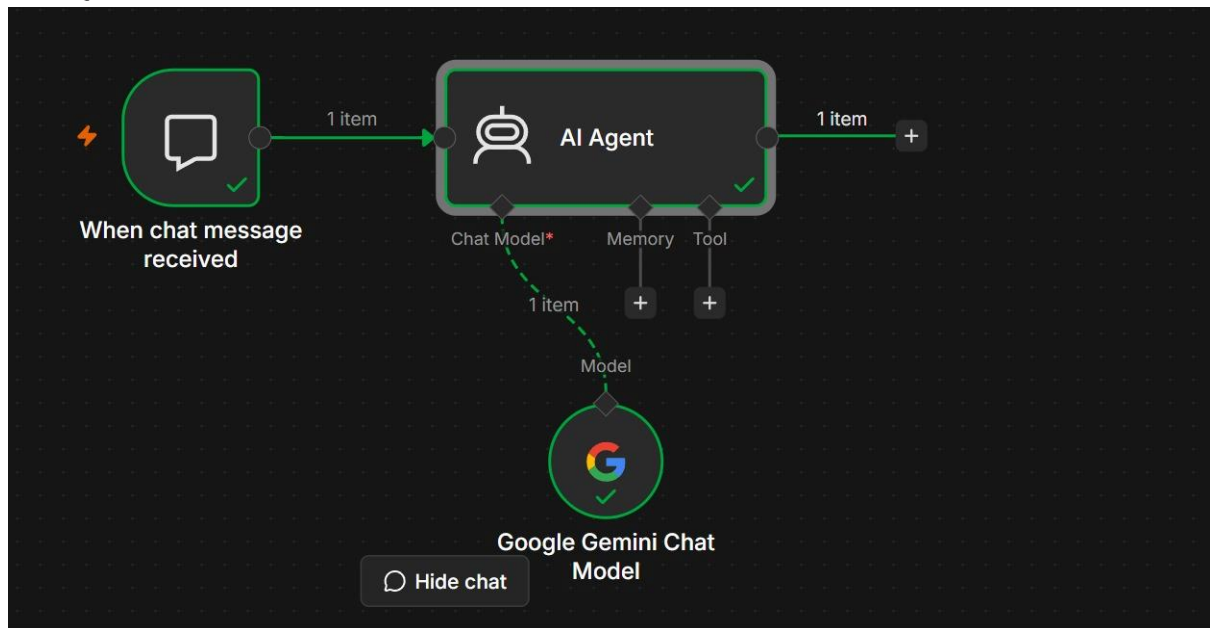


PROJECT-27

AI SQL Generator



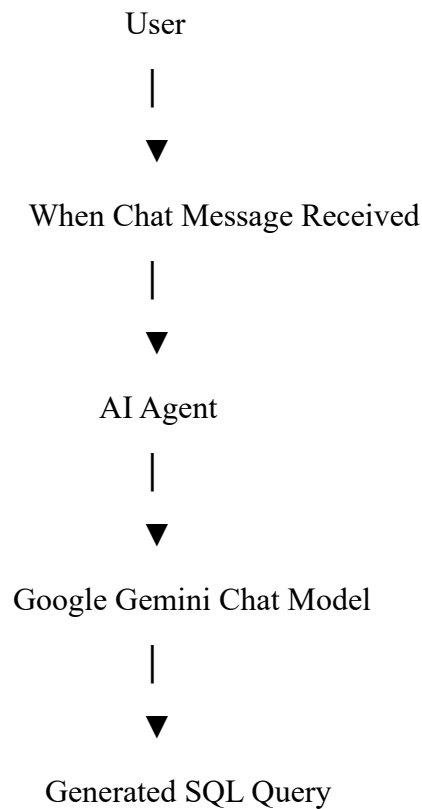
Project Objective

The objective of this project is to build an **AI-powered SQL Generator** that converts natural language into SQL queries. The AI Agent understands the user's database request and generates accurate SQL statements for retrieving, inserting, updating, deleting, and managing data across relational databases.

The AI Agent can generate:

- SELECT queries
- INSERT queries
- UPDATE queries
- DELETE queries
- JOIN queries
- GROUP BY queries
- ORDER BY queries
- Aggregate function queries
- Subqueries
- Database optimization queries

Workflow Architecture



Workflow Components

Step	Node	Purpose
1	When Chat Message Received	Receives the user's database request
2	AI Agent	Interprets the request and prepares the SQL generation prompt
3	Google Gemini Chat Model	Generates the SQL query

Prerequisites

Before building this project, ensure you have:

- n8n (Cloud or Self-hosted)
 - Google AI Studio API Key
 - Google Gemini Chat Model configured
 - Internet connection
-

Step 1 – Create Workflow

Open **n8n**

Click

New Workflow

Rename it

AI SQL Generator

Step 2 – Add Chat Trigger

Add

When Chat Message Received

Purpose

Receives the user's SQL query request.

Example Input

Generate an SQL query to retrieve the names and email addresses of customers who placed more than five orders in the last six months.

Output

```
{  
  "chatInput": "Generate an SQL query to retrieve the names and email addresses of customers  
who placed more than five orders in the last six months."  
}
```

Step 3 – Add AI Agent

Connect

When Chat Message Received

|



AI Agent

Configure

Source for Prompt

Connected Chat Trigger Node

Prompt

```
{{ $json.chatInput }}
```

Step 4 – Configure the System Prompt

Paste the following into the **System Message**.

You are an AI SQL Generator.

Your task is to convert natural language requests into accurate and efficient SQL queries.

Instructions:

- Understand the user's database request.
 - Generate valid SQL syntax.
 - Use meaningful table and column names when the schema is not provided.
 - Generate SELECT, INSERT, UPDATE, DELETE, JOIN, GROUP BY, ORDER BY, HAVING, and subqueries whenever appropriate.
 - Optimize the query for readability and performance.
 - Explain the SQL query briefly after generating it.
 - If the database schema is missing, clearly state any assumptions made.
 - Return only the SQL query followed by the explanation.
-

Step 5 – Add Google Gemini Chat Model

Click

+ **Chat Model**

Choose

Google Gemini Chat Model

Configuration

Setting	Value
Provider	Google Gemini

Setting	Value
Model	Gemini 2.5 Flash (or another available Gemini model)
Authentication	Google AI Studio API Key
Connect it to the AI Agent .	

Final Workflow

When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

Workflow Execution

Step 1

User enters

Generate an SQL query to find the top 10 highest-paid employees.



Step 2

The **When Chat Message Received** node receives the user's request.



Step 3

The **AI Agent** analyzes the request and identifies the SQL operation required.



Step 4

The prompt is sent to the **Google Gemini Chat Model**.



Step 5

Gemini generates an optimized SQL query with a brief explanation.



Step 6

The SQL query is returned to the user.

Sample Inputs and Outputs

Example 1

Input

Generate an SQL query to display all employees earning more than 50,000.

Output

```
SELECT *
```

```
FROM Employees
```

```
WHERE Salary > 50000;
```

Explanation:

This query retrieves all employee records where the salary is greater than 50,000.

Example 2

Input

Generate an SQL query to find the total number of customers in each city.

Output

```
SELECT City, COUNT(*) AS Total_Customers
```

```
FROM Customers
```

```
GROUP BY City;
```

Explanation:

This query groups customers by city and counts the number of customers in each city.

Example 3

Input

Generate an SQL query to display all orders along with customer names.

Output

```
SELECT Customers.CustomerName,  
       Orders.OrderID,  
       Orders.OrderDate  
FROM Customers  
INNER JOIN Orders  
ON Customers.CustomerID = Orders.CustomerID;
```

Explanation:

This query joins the Customers and Orders tables to display customer names with their orders.

Example 4

Input

Generate an SQL query to update the salary of employee ID 101 to 75,000.

Output

```
UPDATE Employees  
SET Salary = 75000  
WHERE EmployeeID = 101;
```

Explanation:

This query updates the salary of employee ID 101.

Workflow Diagram

User

|



When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

|



Generated SQL Query

Error Handling

Error: No Prompt Specified

Cause

The AI Agent did not receive the user's request.

Solution

Verify that the **Prompt** field contains:

```
{{ $json.chatInput }}
```

Error: Authentication Failed

Cause

Invalid Google Gemini API Key.

Solution

Reconnect or update your Google AI Studio credentials.

Error: Too Many Requests

Cause

Gemini API rate limit exceeded.

Solution

- Wait a few minutes.
 - Check your API quota.
 - Retry the request.
-

Learning Outcomes

After completing this project, you will be able to:

- Build an AI-powered SQL Generator in n8n.
 - Configure an AI Agent with Google Gemini.
 - Design prompts for SQL generation.
 - Generate SQL queries from natural language.
 - Build conversational AI workflows for database automation.
-

Real-World Applications

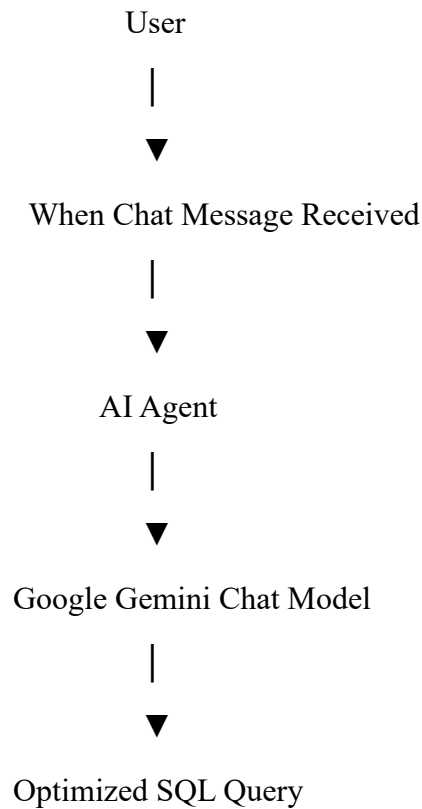
- SQL query generation
 - Database management
 - Data analysis
 - Business intelligence
 - Reporting automation
 - Database learning
 - Developer productivity tools
 - Educational platforms
 - Data engineering
 - AI-powered coding assistants
-

Practice Challenge

Enhance this project by adding:

1. A database type selector (MySQL, PostgreSQL, SQL Server, SQLite, Oracle).
 2. A schema input field to generate schema-specific SQL queries.
 3. An SQL optimization mode to improve query performance.
 4. A Memory node to remember the user's preferred database system.
 5. A PostgreSQL or MySQL node to execute the generated SQL query automatically.
 6. A Google Sheets node to save generated SQL queries for future reference.
-

Final Workflow



Expected Result

The **AI SQL Generator** accepts a natural language database request from the user, uses the **Google Gemini Chat Model** to generate accurate and optimized SQL queries, and returns the SQL statement along with a brief explanation of its functionality. This project teaches prompt engineering, AI-assisted SQL generation, and conversational workflow design in n8n, making it an excellent foundation for building intelligent database automation and developer productivity solutions.